

National University of Computer and Emerging Sciences

AI (CL-461)

Date: April, 22 2026

Course Instructor(s)

Mr. Usman Anwer

Quiz-2

Total Time: 20 mins

Total Marks: 20

Total Questions: 01

Student Name

Roll No

BCS-4F

Section

Student Signature

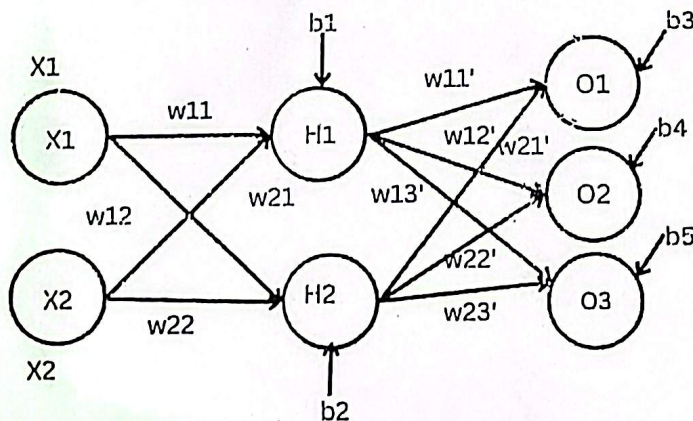
Instructions: Perform all the steps. Weights are typically indexed as w_{ij} , where the first digit i represents the source neuron from the previous layer and the second digit j represents the destination neuron in the current layer

CLO 3: Apply methods of blind as well as informed search on the problem related to the programs.

Question 1:

Q1:

A feed forward neural network is given below:



Initial values are given:

$w_{11}=0.3$	$w_{22}=0.6$	$w_{13'}=0.75$	$w_{23'}=0.90$	target01=1
$w_{12}=0.4$	$w_{11'}=0.65$	$w_{21'}=0.80$	$b_1=b_2=b_3=b_4=b_5=0.$	target02=1.5
$w_{21}=0.5$	$w_{12'}=0.70$	$w_{22'}=0.85$	$X_1=1, X_2=-1$	target03=0.5

Learning rate is 0.5, Apply Gradient Descent Rule to update value of weight w_{22} and bias b_2 only.